

Hani Abusharkh (habusharkh)_PreviewUser Proposal.docx

Document Details

Submission ID

trn:oid::2945:412855372

Submission Date

Sep 14, 2026, 4:43 PM GMT+0

Download Date

Sep 14, 2026, 5:12 PM GMT+0

File Name

Proposal.docx

File Size

37.1 KB

2 Pages

330 Words

1,973 Characters

100% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely revised by AI.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.



10 AI-generated 100%

Likely generated or likely generated and revised by AI.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely human generated as AI generated and likely AI generated as human generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Research Proposal

Reconstructed Jira histories for late sprint closure and unfinished committed scope

Background

Sprint Planning is a natural point for early warning because the team can still adjust scope before execution begins. Repository data may contain useful signals, but only if the features reflect what was known at that time.

Research aim

The study will examine whether Jira history can rank two repository-derived outcomes: late sprint closure and unfinished scope that was committed at sprint start. The work will also test whether simple scope measures explain most of the predictable signal.

Research questions

1. How much ranking signal is present for late closure and unfinished committed scope?
2. Does a transparent scope-only model perform as well as more flexible models?
3. How stable are the results across time and across projects?
4. How sensitive are the findings to missing historical provenance and leakage-prone evaluation choices?

Data

The planned dataset is TAWOS, a relational collection of public Jira projects. The working subset contains eight projects. Sprint, Issue, and Change_Log records are used so that selected fields can be reconstructed at the start of each sprint.

Method

Sprint membership and Story Points will be reconstructed from timestamped change history. Nonclosed sprints will be removed before labels are built. Models will include logistic regression, random forest, and XGBoost, alongside prevalence and scope-only baselines. Evaluation will use forward temporal splits and held-out-project transfer.

Evaluation

ROC-AUC will be the main ranking measure. F1, precision, recall, PR-AUC, and Brier score will be reported with their limitations. Project-level uncertainty, multiple temporal cut points, provenance sensitivity, feature permutation, and leakage demonstrations will be used to test the robustness of the main conclusions.

Expected contribution

The intended contribution is methodological as well as predictive: a reproducible example of how timing, scope provenance, prevalence, project heterogeneity, and leakage affect sprint-analytics results.

Limits

The study is retrospective and uses open-source repositories. It will not claim that a risk score improves Scrum decisions; that would require prospective validation with real teams.